

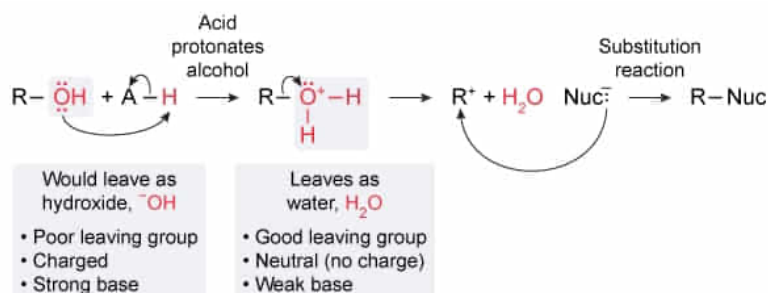
Hydroxyl substituents in alcohols are poor leaving groups, and are not readily eliminated during substitution reactions. Which of the following modifications would make the hydroxyl substituent of an alcohol a better leaving group?

- ☐ A. Deprotonation [7%]
- ✓ ☒ B. Addition of an acid [77%]
- ☐ C. Addition of a strong base [8%]
- ☐ D. Conversion to an acetal [6%]

Correct

77% answered correctly

Explanation:



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Substitution reactions require a nucleophile and an electrophile, as well as a good leaving group. A bond is formed between the nucleophile and electrophile, and the bond between the electrophile and leaving group is broken. The leaving group gains a pair of electrons when its bond with the electrophile is broken, and the leaving group is more likely to depart if the electrons gained can form a stable arrangement. Weak bases make better leaving groups than strong bases because they are often neutral molecules or have delocalized charge. Neutral species, such as water, formed after leaving make good leaving groups because uncharged species are more stable than charged species.

Alcohols must be modified to form a good leaving group for substitution reactions because hydroxide (OH^-) is a strong base. The **addition of acid** to an **alcohol** causes the hydroxyl group to become **protonated**, forming bound water with a positive charge on oxygen. Water is a weak base, and becomes neutral once it leaves. Therefore, **protonation** of an alcohol forms a **good leaving group**.

(Choices A and C) The addition of a strong base will cause deprotonation of the alcohol, creating a negative charge on the oxygen atom of the alcohol. Atoms with a negative charge tend to donate electrons and act as a nucleophile rather than accept electrons and act as a **leaving group**.

(Choice D) **Acetals** are carbonyl protecting groups formed by the addition of an alcohol to a ketone or aldehyde in acidic conditions. Protecting groups render a functional group unreactive, and therefore would not make alcohols better leaving groups.

Educational objective:

Hydroxyl groups in alcohols are poor leaving groups because hydroxide is a strong base. Hydroxide must be converted to a good leaving group to participate in substitution reactions. The addition of acid to an alcohol protonates the hydroxyl group, forming a bound, positively charged water molecule. Water is a weak base and forms a neutral molecule upon leaving, which makes it a good leaving group.

Time spent:0 sec

Subject:
Organic Chemistry

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System:
Functional Groups and Their Reactions